Lagrangian Variable Cosmology, final-v1.4: Observational Stress Test of the v1.1 Modulation under Perturbative Completion

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Abstract

Working paper v1.1 of this series exhibited an exact half-twist sine–Gordon kink whose squared quarter-angle reproduces the observed late-time modulation profile $T(z) - 1 = A\,\mathrm{sech}^2((\theta - \theta_c)/w)$ in the log-time coordinate $\theta = \ln(1 + z)$, with $\theta_c = \ln\varphi$, $w = (\ln\varphi)^3$, $A = (\ln\varphi)^4$. The construction was static and one-dimensional. This paper asks a separate question: does the v1.1 modulation remain observationally viable when the same coordinate function $\vartheta_{\rm kink}(z)$ is fed into a non-minimal coupling $f(\vartheta) = \exp[-\alpha(1 - \cos(\vartheta/2))]$ and lifted to the Friedmann and growth equations? The kink identity $1 - \cos(\vartheta_{\rm kink}/2) = 2\,\mathrm{sech}^2((\theta - \theta_c)/w)$ makes $f(\vartheta_{\rm kink}(z))$ a closed function of z, and the modified Friedmann equation returns $T(z) = \exp[\alpha_b\,\mathrm{sech}^2((\theta - \theta_c)/w)]$ with the same lock parameters. The construction reduces to ΛCDM at numerical precision for $z \gtrsim 30$ without imposed restriction. On the PP-excluded data set ($N = 37$) the locked construction with $\alpha_b = A$ returns $\chi^2 = 56.090$ at $k = 2$, against $\chi^2 = 78.962$ at $k = 3$ for ΛCDM, giving $\Delta\mathrm{BIC} = -26.48$. On the PP-included data set ($N = 1738$) the same locks return $\chi^2 = 1601.964$ at $k = 2$ against $\chi^2 = 1624.848$ at $k = 3$ for ΛCDM, giving $\Delta\mathrm{BIC} = -30.34$. A free-$\alpha_b$ fit returns $\alpha_b = 0.0542$, within 1.2% of the lock value $A = (\ln\varphi)^4 = 0.0536$. The growth equation with $G_{\rm eff}/G = 1/f(\vartheta_{\rm kink})$ absorbs the v1.3 non-minimal coupling channel; the $f\sigma_8$ data set ($N = 16$) returns $\alpha_g = 0.107$ at $\sigma_8(z = 0) = 0.813$, and the implied $S_8 = 0.783$ places the construction within 0.5σ of the weak-lensing combined central value 0.778 ± 0.010. The construction is offered as a re-derivation of the v1.1 profile in a form that is exact at all redshifts and that absorbs the v1.3 growth coupling into the same action; it does not extend the phenomenology and does not select the lock values.

1 Introduction

The phenomenological status of a localised excursion in the late-time expansion-rate modulation $T(z) = H(z)/H_{\Lambda\rm CDM}(z)$ is recorded in working papers v0.1–v0.3 and v16 [1, 2, 3, 4]. Working paper v1.1 [5] exhibited the modulation as the squared quarter-angle of a half-twist sine–Gordon kink in the $(1 + 0)$-dimensional sense; working paper v1.3 [6] added a non-minimal coupling to address the growth-rate data. Both papers treated $\vartheta(\theta)$ as a static profile in the coordinate $\theta = \ln(1 + z)$, with no equation of motion on an FLRW background.

This paper records what happens when the same coordinate function $\vartheta_{\rm kink}(z)$ is substituted into the non-minimal coupling $f(\vartheta) = \exp[-\alpha(1 - \cos(\vartheta/2))]$ and the coupling is allowed to enter the Friedmann equation as well as the growth equation. The question addressed is whether the v1.1 modulation, lifted in this way, remains observationally viable across BAO, supernovae, distance priors, and growth-rate data, and what the lifted form implies at high redshift.

Section 2 records the v1.1 baseline in the form used by the rest of this paper. Section 3 records the observational status of the v1.1 baseline across H_0, σ_8, BAO, and CMB-distance data. Section 4 records the perturbative completion through (8)–(11) and the resulting fits. Section 5 states what the construction does and does not select. Appendices A and B record the lock-selection question and the status of the static substitution.

2 Baseline

Working paper v1.1 fits the multiplicative modulation $H(z) = H_{\Lambda\text{CDM}}(z; H_0, \Omega_m)\, T(z)$ with

$$T(z) \;=\; 1 \;+\; A\,\operatorname{sech}^2\!\big((\theta-\theta_c)/w\big), \qquad \theta = \ln(1+z), \tag{1}$$

and the lock pattern

$$\theta_c = \ln\varphi, \quad A = (\ln\varphi)^4, \quad w = (\ln\varphi)^3, \quad \Omega_m = 1/(1+\exp(3\ln\varphi/\varphi)) = 0.290653. \tag{2}$$

The profile (1) is the squared quarter-angle of the static kink $\vartheta_{\text{kink}}(\theta) = 8\arctan(\exp[(\theta-\theta_c)/w])$ of the half-twist sine–Gordon potential $V(\vartheta) = m^2[1-\cos(\vartheta/2)]$ at $m = 2/w$; the identity used is

$$\sin^2\!\big(\vartheta_{\text{kink}}(\theta)/4\big) \;=\; \operatorname{sech}^2\!\big((\theta-\theta_c)/w\big). \tag{3}$$

Working paper v1.3 retains the static profile and adds a non-minimal coupling

$$f(\vartheta) \;=\; \exp\!\big[-\alpha\big(1-\cos(\vartheta/2)\big)\big], \qquad G_{\text{eff}}/G \;=\; 1/f(\vartheta_{\text{kink}}), \tag{4}$$

fit at the locked single-rung value $\alpha = 0.2258$ to recover $\sigma_8 = 0.811$ at Ω_m given by (2).

The lock pattern (2) is imposed as input in v1.1 and is retained throughout this paper. The selection principle that picks $\theta_c = \ln\varphi$ and $A = (\ln\varphi)^4$ remains open; this is discussed in Appendix A.

3 Observational status of the v1.1 baseline

The locked construction (1)–(2), fit on the PP-included data set ($N = 1738$, comprising Pantheon+ unbinned supernovae, DESI DR1/DR2 BAO, BOSS DR12, eBOSS DR16, Planck 2018 distance priors, and SH0ES H_0), returns $\chi^2 = 1601.999$ at $k = 2$ against $\chi^2 = 1624.848$ at $k = 3$ for ΛCDM (v1.1, §3.3). Table 1 records the status of the same construction across four observational categories that have appeared in the tension literature since v1.1 was written.

Table 1: Observational status of the v1.1 locked construction (1)–(2) across four data categories, as recorded on the data sets available at the time of this paper.

Category	v1.1 baseline status	Reference
BAO (DESI DR1+DR2, BOSS, eBOSS)	$\Delta\chi^2_{\text{BAO}} \approx -10$ vs. ΛCDM free	v1.1, §3.3
SN distance scale (Pantheon+)	comparable to ΛCDM free	v1.1, §3.3
H_0	best-fit 69.06 between SH0ES and Planck	v1.1, §3.3
σ_8 from background-only $T(z)$	$\sigma_8(0) = 0.803$ (below Planck 0.811)	v1.3, §11
$S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ (weak lensing)	$S_8 = 0.781$ at v1.1 Ω_m-lock	this paper
CMB distance prior (R, ℓ_A)	$\chi^2_{\text{DP}} \approx 12$ (ΛCDM free: 22)	this paper

The baseline returns acceptable values across the four categories. The H_0 value sits between SH0ES and Planck; the σ_8 value sits below Planck. Working paper v1.3 records the σ_8 value as the motivation for the non-minimal coupling (4). The present paper records two further observations on the baseline.

High-redshift extrapolation. The static profile (1) satisfies $T(z)-1 \to 0$ as $z \to \infty$, but the construction in v1.1 is not derived from an action on the FLRW background; its extrapolation to recombination ($z \approx 1090$) and BBN ($z \approx 10^9$) is taken without justification in v1.1. The lifted form derived in Section 4 returns the same envelope as the limit of a Friedmann-equation modification and provides the missing extrapolation.

Two non-minimal coupling channels. The v1.3 coupling (4) enters only the growth equation through $G_{\text{eff}}/G = 1/f$. The same coupling would in principle enter the Friedmann equation through the Jordan-frame factor $f(\vartheta)R/2$. v1.3 does not address the background channel; the present paper makes that channel explicit and records the consequence.

These observations motivate the perturbative completion of Section 4.

4 Perturbative completion

4.1 Lift of the v1.1 profile through a non-minimal coupling

Consider the Jordan-frame action

$$S = \int d^4x\,\sqrt{-g}\left[\tfrac{1}{2}\,f(\vartheta)\,R - \tfrac{1}{2}(\partial\vartheta)^2 - V(\vartheta)\right] + S_{\text{matter}}[g_{\mu\nu},\psi], \tag{5}$$

with f and V as in v1.3. We do not solve the equation of motion for ϑ in (5); we substitute the v1.1 coordinate function $\vartheta_{\text{kink}}(z)$ defined by setting $\theta = \ln(1+z)$ in the v1.1 expression, and treat $f(\vartheta_{\text{kink}}(z))$ as a fixed external function of z. The status of the substitution is recorded in Appendix B.

From the v1.1 kink identity (3) and the trigonometric relation $1-\cos(x) = 2\sin^2(x/2)$ one has

$$1-\cos\!\big(\vartheta_{\text{kink}}(\theta)/2\big) = 2\,\operatorname{sech}^2\!\big((\theta-\theta_c)/w\big), \tag{6}$$

so that

$$f(\vartheta_{\text{kink}}(z)) = \exp\!\big[-2\alpha\,\operatorname{sech}^2((\theta-\theta_c)/w)\big], \qquad \theta = \ln(1+z). \tag{7}$$

With $f(\vartheta_{\text{kink}}(z))$ a closed function of z, and neglecting the kinetic and potential contributions of ϑ to the background (Appendix B), the $(0,0)$ Einstein equation reduces to

$$3\,H^2(z)\,f(\vartheta_{\text{kink}}(z)) = 3\,H^2_{\Lambda\text{CDM}}(z), \tag{8}$$

where $H_{\Lambda\text{CDM}}$ is the standard ΛCDM expansion rate at the same parameters. Combining with (6) gives

$$\boxed{T(z) \equiv H(z)/H_{\Lambda\text{CDM}}(z) = \exp\!\big[\,\alpha_b\,\operatorname{sech}^2\!\big((\theta-\theta_c)/w\big)\,\big], \quad \theta = \ln(1+z),} \tag{9}$$

with the same θ_c, w as in v1.1. The amplitude parameter α_b is the background-channel coupling. The small-α_b expansion of (9) returns the v1.1 form $T \approx 1 + \alpha_b \operatorname{sech}^2$ at leading order under the identification $\alpha_b = A$; exact peak-amplitude matching gives $\alpha_b = \ln(1+A) = 0.05223$, 2.6% below the lock value $A = 0.05362$.

The growth equation reads

$$D''(N) + \big(2 + d\ln H/dN\big)\,D'(N) = \tfrac{3}{2}\,\Omega_m(a)\,\frac{G_{\text{eff}}}{G}\,D(N), \qquad N = \ln a, \tag{10}$$

with $\Omega_m(a) = \Omega_m a^{-3}/(H/H_0)^2$ evaluated using the modulated H of (9), and

$$\frac{G_{\text{eff}}}{G} = \frac{1}{f(\vartheta_{\text{kink}}(z))} = \exp\!\big[+2\,\alpha_g\,\operatorname{sech}^2((\theta-\theta_c)/w)\big], \tag{11}$$

where α_g is the growth-channel coupling. We keep α_b and α_g as independent fit parameters in this paper and discuss the constraint between them in Section 5.

4.2 High-redshift behaviour

Table 2: Magnitude of $T(z) - 1$ from (9) at $\alpha_b = A$. "Underflow" denotes values below double-precision floating-point representation.

z	epoch	$T(z) - 1$
0	today	3.81×10^{-5}
0.618	peak ($z = 1/\varphi$)	5.51×10^{-2}
1	mid-z BAO	4.58×10^{-3}
2.33	Lyα BAO	5.07×10^{-7}
10	high-z	2.22×10^{-16}
30	–	$< 10^{-30}$
1090	recombination	underflow
10^9	nucleosynthesis	underflow

The exponential-envelope form (9) reduces to ΛCDM at double-precision floating-point precision for $z \gtrsim 30$. Recombination and BBN are therefore unaffected at the background-Hubble level within the floating-point precision used in the fits. This is a structural property of (9) and is unchanged by the lock values: any α_b of order unity returns the same conclusion at these redshifts.

4.3 Background fits

The fits use the v1.3 likelihood pipeline [6] unchanged. ΛCDM is fit with three free parameters $H_0, \Omega_m, \Omega_b h^2$. The v1.1 sech^2 profile and the rewriting (9) are fit with the four locks of (2), leaving H_0 and $\Omega_b h^2$ as the two free parameters ($k = 2$); α_b is either fixed at the lock $\alpha_b = A$ or freed as a third parameter. Tables 3 and 4 record the results; Table 5 records the distance-prior contribution separately.

Table 3: PP-excluded data set, $N = 37$. Columns: number of free parameters k, best-fit H_0 in km s^{-1} Mpc^{-1}, χ^2, and ΔBIC relative to ΛCDM. The v1.1 row reproduces v1.1, §3.2.

Model	k	H_0	χ^2	ΔBIC
ΛCDM	3	70.34	78.962	0
v1.1 sech^2, locks	2	69.04	56.134	−26.44
v1.4 Eq. (9), $\alpha_b = A$	2	69.04	56.090	−26.48
v1.4 Eq. (9), α_b free	3	69.03	56.086	−22.88

The free-α_b fit on the PP-excluded data set returns $\alpha_b = 0.0541$, 0.9% above the lock value $A = (\ln \varphi)^4 = 0.0536$.

Table 4: PP-included data set, $N = 1738$. The v1.1 row reproduces v1.1, §3.3.

Model	k	H_0	χ^2	ΔBIC
ΛCDM	3	70.40	1624.848	0
v1.1 sech^2, locks	2	69.06	1601.999	−30.31
v1.4 Eq. (9), $\alpha_b = A$	2	69.06	1601.964	−30.34
v1.4 Eq. (9), α_b free	3	69.06	1601.958	−22.89

On the PP-included data set the free-α_b fit returns $\alpha_b = 0.0542$, 1.2% above the lock value. The χ^2 difference between v1.1 sech^2 and the rewriting (9) at the same locks is $|\Delta\chi^2| < 0.05$ on both data sets; the two profiles are not distinguished by the present data.

Table 5: Planck 2018 distance-prior χ^2 contribution at the PP-included best-fit parameters. The Planck DP central values are $R = 1.7493 \pm 0.0049$, $\ell_A = 301.462 \pm 0.090$, $\Omega_b h^2 = 0.02236 \pm 0.00015$.

Model	H_0	$\chi^2_{\rm DP}$	R	ℓ_A
ΛCDM	70.40	21.86	1.7265	301.249
v1.1 sech^2	69.06	12.15	1.7325	301.287
v1.4 Eq. (9)	69.06	12.28	1.7324	301.289

Both modulated constructions return a $\chi^2_{\rm DP}$ contribution roughly 9.7 below ΛCDM at the same data; this difference is a consequence of the H_0, Ω_m shift in the modulated fits. Full-Boltzmann CMB likelihoods (CAMB/CLASS) have not been evaluated and are noted in Section 5.

4.4 Growth-rate fit

The $f\sigma_8(z)$ data set used here consists of $N = 16$ measurements spanning $z \in [0.067, 1.491]$: 6dFGS [11], SDSS-MGS [12], BOSS DR12 [13], WiggleZ [14], eBOSS LRG/ELG/QSO [15], and DESI DR2 BGS/LRG/ELG/QSO [16]. Table 6 records the fits.

Table 6: $f\sigma_8$ direct fit, $N = 16$ data points. Parameters held locked: $\theta_c = \ln\varphi$, $w = (\ln\varphi)^3$, $\Omega_m = 0.290653$. $\sigma_8(z = 0)$ given relative to Planck primordial normalisation $\sigma_8^{\rm Planck} = 0.811$.

Model	k	χ^2	$\sigma_8(z = 0)$
ΛCDM (Ω_m at v1.1 lock)	0	12.181	0.8110
v1.4, $\alpha_b = A$, $\alpha_g = 0$	0	13.776	0.8028
v1.4, $\alpha_b = A$, α_g free	1	12.243	0.8130
v1.4, α_b, α_g free	2	11.998	0.8118

The α_g-free fit at fixed $\alpha_b = A$ returns $\alpha_g = 0.107$, below the v1.3 single-rung value $\alpha = 0.226$ fit on σ_8 alone [6]. The residual root-mean-square pull on the 16 points is 0.88; no point exceeds 2σ in absolute pull.

4.5 Implied S_8

Table 7: Implied $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$ at the v1.4 lock, together with three measurement groups. Errors of the WL combined entry follow the inverse-variance combination of DES Y3, KiDS-1000, HSC Y3, and DES+KiDS [17, 18, 19, 20].

Construction or measurement	$\sigma_8(z = 0)$	S_8	to WL avg.
ΛCDM ($\Omega_m = 0.315$, Planck normalisation)	0.811	0.831	$+5.3\sigma$
ΛCDM (Ω_m at v1.1 lock = 0.291)	0.793	0.781	$+0.3\sigma$
v1.4, $\alpha_b = A$, $\alpha_g = 0$	0.785	0.773	-0.5σ
v1.4, $\alpha_b = A$, $\alpha_g = 0.107$	0.795	0.783	$+0.5\sigma$
v1.4, $\alpha_b = A$, $\alpha_g = 0.226$	0.808	0.796	$+1.8\sigma$
WL combined (DES Y3, KiDS-1000, HSC Y3, DES+KiDS)		0.778 ± 0.010	—
CMB primary (Planck 2018)	0.811	0.832 ± 0.013	$+4.5\sigma$
CMB lensing (ACT DR6 + Planck 2018 lensing)		0.818 ± 0.015	$+3.0\sigma$

The S_8 value from the v1.4 construction at α_g between 0 and 0.107 lies within 0.5σ of the WL-combined central value 0.778 ± 0.010. The same construction lies 3–5σ from the CMB-primary central value 0.832 ± 0.013. The Ω_m-lock value $\Omega_m = 0.2907$ from (2) accounts for most of the S_8 shift in Table 7: ΛCDM at this Ω_m already returns $S_8 = 0.781$. The modulation (9) contributes a further -0.008 at $\alpha_g = 0$ and $+0.002$ at $\alpha_g = 0.107$.

5 Interpretation

The construction (9)–(11) provides a route to the v1.1 modulation profile that is exact at all redshifts within the floating-point precision of the fits, and that incorporates the v1.3 non-minimal coupling channel into the same action (5). Tables 3 and 4 record that the rewriting returns the v1.1 χ^2 to within 0.05 on both data sets, and that the free-α_b best-fit recovers the algebraic lock $A = (\ln\varphi)^4$ from the data to within 1.2%.

Three observations follow from the fits.

Two coupling channels share one function. Equations (9) and (11) derive from the same $f(\vartheta) = \exp[-\alpha(1-\cos(\vartheta/2))]$ in (5), but enter background and growth channels through different combinations. A first-principles derivation that reduces $\{\alpha_b, \alpha_g\}$ to a single parameter requires the metric variation of (5) including the kinetic contribution of ϑ; this is not undertaken here. Empirically, the fits return $\alpha_b \approx A$ (from background fits, Tables 3 and 4) and $\alpha_g \approx 0.107$ (from $f\sigma_8$ at fixed $\alpha_b = A$, Table 6). Whether the action (5) with a consistent treatment forces a relation between the two values is open.

Position across current tensions. The locked best-fit $H_0 \approx 69.06$ on both background data sets lies between Planck 2018 [8] and SH0ES [10]. The implied $S_8 \approx 0.78$ lies within 0.5σ of the weak-lensing combined central value and 3–5σ from the Planck primary CMB central value. These two values are placed where the construction sits; no claim of tension resolution is attached. The construction provides a single locked structure under which both observations appear at their fitted values.

Outstanding evaluations. Tables 4 and 5 use the compressed Planck distance-prior likelihood of [7]. A full Boltzmann calculation (CAMB or CLASS) with the modulation (9) substituted into the background expansion, and the corresponding modification of the late-time integrated Sachs–Wolfe contribution, has not been performed and is recorded as outstanding.

A Lock selection

The lock pattern (2) is imposed as input in v1.1, in v1.3, and in this paper. The free-fit values that motivate the lock are recorded in v0.3 [3]: a centre $\theta_c = 0.4812$ matching $\ln\varphi$ to four decimal places, and a width $w \approx 0.110$ close to both $(\ln\varphi)^3 = 0.11143$ and $2/\varphi^6 = 0.11146$, which differ from each other by 0.02% and are not distinguished by present data (v1.1, §2). The amplitude $A = (\ln\varphi)^4$ enters in v1.1 as the parameter ε^2 identified with the squared quarter-angle prefactor.

The free-α_b fits of Tables 3 and 4 return $\alpha_b = 0.0541$ and $\alpha_b = 0.0542$ on the two data sets, within 1.2% of $A = (\ln\varphi)^4 = 0.05362$. This is recorded as a data-driven coincidence with the v1.1 algebraic lock, not as a derivation of the lock. A selection principle that picks the lock values from a larger structural argument has not been exhibited in this paper or in v1.1–v1.3 and remains the substantive open question of the programme.

B Status of the substitution $\vartheta = \vartheta_{\rm kink}(z)$

The substitution $\vartheta = \vartheta_{\rm kink}(z)$ in (8) treats the v1.1 coordinate function as a fixed external z-profile rather than as the solution of the equation of motion that follows from (5) on an FLRW background. Numerical integration of that equation with the v1.1 mass $m = 2/(\ln\varphi)^3 \approx 17.95$ on an ΛCDM background, with initial conditions in the basin of attraction of the potential minimum at $\vartheta = 4\pi$, does not yield a trajectory that crosses to an adjacent minimum at $z \approx 1/\varphi$; we tested initial conditions spanning $|\vartheta_{\rm init} - 4\pi| \in [10^{-6}, 2]$ and $\dot{\vartheta}_{\rm init} \in [0, 20]$. The substitution is therefore a re-use of the v1.1 static profile, not the classical FLRW evolution of the scalar field in (5).

The kinetic and potential contributions of $\vartheta_{\rm kink}(z)$, treated as a coordinate function, are dropped in (8). The retained term $f(\vartheta_{\rm kink}(z))R/2$ gives the closed expression (9). Whether a self-consistent dynamical treatment can be constructed in which the profile (9) is preserved is open.

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